MATH221 & CE122 : Differential Equations & Numerical Methods & Applied Programming

Project: Study of the application of ODE models and numerical methods in investigating a real-world problem

Project goal: Each group of students will be responsible for identifying a study that implemented ODE and numerical methods to investigate a real-world problem. Some examples of such studies are cancer therapy, oil spills, malaria, climate change, etc. You will be required to understand the model, investigate different aspects of models, such as parameter sensitivity and/or implement a modification in the model (for the differential equations course), **and** investigate different numerical methods, perform an error analysis and output (figures, animations, etc.) will be required for the Applied Programming course.

The outputs will be presented on a website. You can use Wix, Square Spaces or other web servers.

Parts of the website:

The website should contain the following sections:

For the Differential Equations course

- Introduction of the topic of your project
- Description of the model with a clear explanation of the equations and how they were derived.
- Discussion of the results of the study
- A creative picture depiction of the project. An example can be seen here.
- A section that presents the modifications that you are going to study and implement in your model
- A section that presents and discusses your results
- A section on what further research

For the Applied Programming course:

- Explanation of the code
- Comparison of the approximating the solution using different solvers
- Error analysis between the different approximations
- Different approaches for the model outputs etc. figures, animations etc.

After you have completed most of the project, create a short 2-minute promo video of the project; here are some examples for reference.

Thus, in conclusion, you have to create a MATLAB script to approximate the solution using an ODE solver. If you can solve the equation analytically, please do so and include that in your results section. Once a script has been developed, then changes in the values of the parameter(s) can be investigated, and their effect on the solution of the equation can be analysed. The values given to the parameters should be justified, either by references or by doing a sensitivity analysis (varying the parameter(s) through a clear and organised set of values). You need also to investigate the effect of using different solvers on the solution of the ODE; however, be aware that these changes might not be visible for a specific parameter value and so a sensitivity analysis might also be required. ***Ideally, if possible, try to modify the existing model to investigate a similar scenario.***

Submission Requirements: This project will be presented as a website. Depending on the time you want to devote and your expertise, you can create the website from scratch and find a hosting domain (Google Drive does allow for this, https://www.labnol.org/internet/host-website-on-google-drive/28178/#:~:text=You%20can%20use%20Google%20Drive,audio%20%26%20video%20files%20including%20podcast) or you can use a platform such as Wix which would host your website. Although these are websites presenting projects on different topics some not related to mathematical modelling, they can serve as inspirations:

1. https://2020.igem.org/Team:William_and_Mary
2. https://sarfomirriam20.wixsite.com/modelling-forgetfuln
3. https://sirsunantsalot.wixsite.com/my-site-5
4. https://deborahdebre.wixsite.com/malariaode
5. https://tesla3003.wixsite.com/diff-eq-project
6. https://edithboakye.wixsite.com/odeproject

There will be peer evaluation of the projects in addition to the instructor's assessment, evaluation rubric and assignments will follow.